Physical Chemistry (I)

Lecture 22

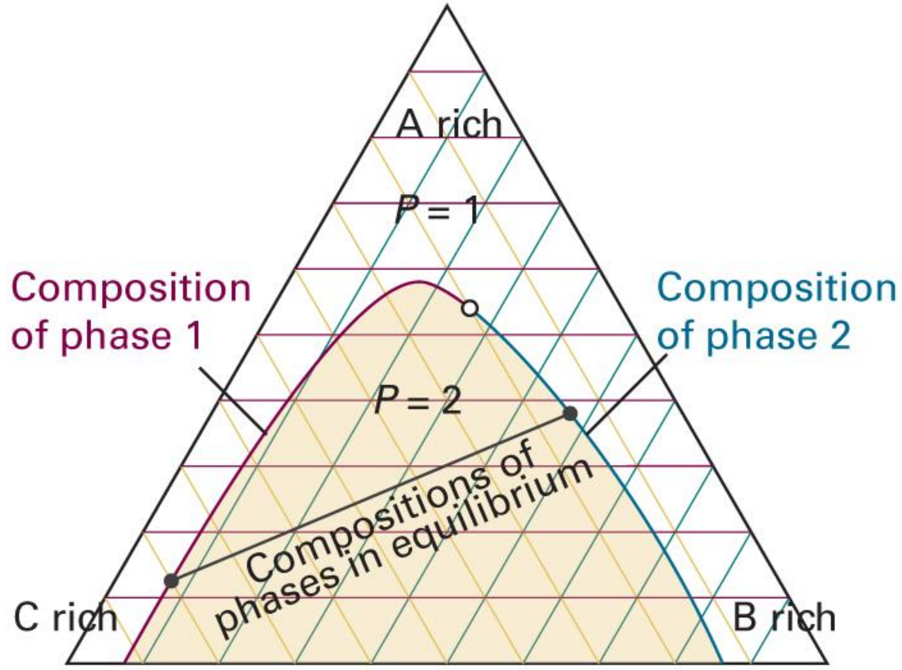
Electrochemical Cells

In the Last Class

- Phase diagrams of ternary systems
- Chemical reactions and equilibria
  - Reaction quotient and equilibrium constant
  - Activity approximations
  - Gaseous reactions
- Le Chatelier's principle

Last class

Ternary Phase Diagram



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.5)

Last class

Chemical Reaction

For an arbitrary reaction,

$$\sum_{\text{reactant } i} \nu_i \{i\} \rightleftarrows \sum_{\text{product } j} \nu_j \{j\}$$

e.g. 2A + B ⇄ 3C + D

$\nu_{\mathrm{A}} = 2$, $\nu_{\mathrm{B}} = 1$, $\nu_{\mathrm{C}} = 3$, $\nu_{\mathrm{D}} = 1$

Now,

$$\mathrm{d}G = \sum_{\text{product } j} \mu_j \mathrm{d}n_j + \sum_{\text{reactant } i} \mu_i \mathrm{d}n_i$$

Extent of Reaction

$$\mathrm{d}G = \sum\nolimits_{\text{product } j} \mu_j \mathrm{d}n_j + \sum\nolimits_{\text{reactant } i} \mu_i \mathrm{d}n_i$$

$$\mathrm{d}G = \left(\sum\nolimits_{\text{product } j} \mu_j \nu_j - \sum\nolimits_{\text{reactant } i} \mu_i \nu_i \right) \mathrm{d}\xi$$

$$\Delta_\mathrm{r} G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \sum\nolimits_{\text{product } j} \mu_j \nu_j - \sum\nolimits_{\text{reactant } i} \mu_i \nu_i$$

Activity

$$\Delta_{\mathrm{r}}G = \sum\nolimits_{\mathrm{product}\ j} \nu_j \mu_j - \sum\nolimits_{\mathrm{reactant}\ i} \nu_i \mu_i$$

For each species x,

$$\nu_x \mu_x = \nu_x(\mu_x^{\ominus} + RT\ln a_x) = \nu_x \mu_x^{\ominus} + RT\ln\left(a_x^{\nu_x}\right)$$

and one can show

$$\Delta_{\mathrm{r}}G = \Delta_{\mathrm{r}}G^{\ominus} + RT\ln\left(\prod_{\mathrm{product}\ j} a_j^{\nu_j} \Big/ \prod_{\mathrm{reactant}\ i} a_i^{\nu_i}\right)$$

Q and K

$$\Delta_\mathrm{r}G = \Delta_\mathrm{r}G^\ominus + RT\ln Q$$

where the reaction quotient Q is defined as

$$Q = \frac{\prod_{\mathrm{product}\ j} a_j^{\nu_j}}{\prod_{\mathrm{reactant}\ i} a_i^{\nu_i}}$$

At equilibrium, $\Delta_\mathrm{r}G = 0$

$$0 = \Delta_\mathrm{r}G^\ominus + RT\ln Q_\mathrm{eq} = \Delta_\mathrm{r}G^\ominus + RT\ln K$$

where the equilibrium constant K is Q at equilibrium

Last class

Approximation for Activity

State	Measure	Approximation for a_J	Definition
Solute	molality	$b_J/b^\ominus$	$b^\ominus = 1\,\mathrm{mol\,kg^{-1}}$
	molar concentration	$[J]/c^\ominus$	$c^\ominus = 1\,\mathrm{mol\,dm^{-3}}$
Gas phase	partial pressure	$p_J/p^\ominus$	$p^\ominus = 1\,\mathrm{bar}$
Pure solid, liquid		1 (exact)	

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., p. 210)

Equilibrium Constants

For gaseous reactions (assuming perfect gases),

$$K_p = \frac{\prod_{\text{product } j}\left(p_j/p^{\ominus}\right)^{\nu_j}}{\prod_{\text{reactant } i}\left(p_i/p^{\ominus}\right)^{\nu_j}}$$

$$K_c = \frac{\prod_{\text{product } j}\left([j]/c^{\ominus}\right)^{\nu_j}}{\prod_{\text{reactant } i}\left([i]/c^{\ominus}\right)^{\nu_i}}$$

$$K_p = K_c\left(\frac{c^{\ominus}RT}{p^{\ominus}}\right)^{\Delta\nu}$$

Last class

Theoretical Estimation

For a perfect-gas reaction,

$$\mu_x = -RT \ln\left(\frac{z_x}{N_x}\right)$$

and one can show

$$K_c = \frac{\prod_{\text{product } j}([j]/c^{\ominus})^{\nu_j}}{\prod_{\text{reactant } i}([i]/c^{\ominus})^{\nu_i}} = \frac{\prod_{\text{product } j}\{z_j/(N_\text{A} V c^{\ominus})\}^{\nu_j}}{\prod_{\text{reactant } i}\{z_i/(N_\text{A} V c^{\ominus})\}^{\nu_i}}$$

Last class

Le Chatelier's Principle

When subjected to a disturbance,

a system at equilibrium tends to respond in a way

that minimizes the effect of the disturbance.

Disturbances to consider:

- Amount
- Volume & pressure (for a gaseous reaction)
- Temperature

van't Hoff Equation

$$\ln K = -\Delta_r G^{\ominus}/RT$$

Using the Gibbs-Helmholtz equation,

$$\frac{\mathrm{d}\ln K}{\mathrm{d}T} = \frac{\Delta_r H^{\ominus}}{RT^2}$$

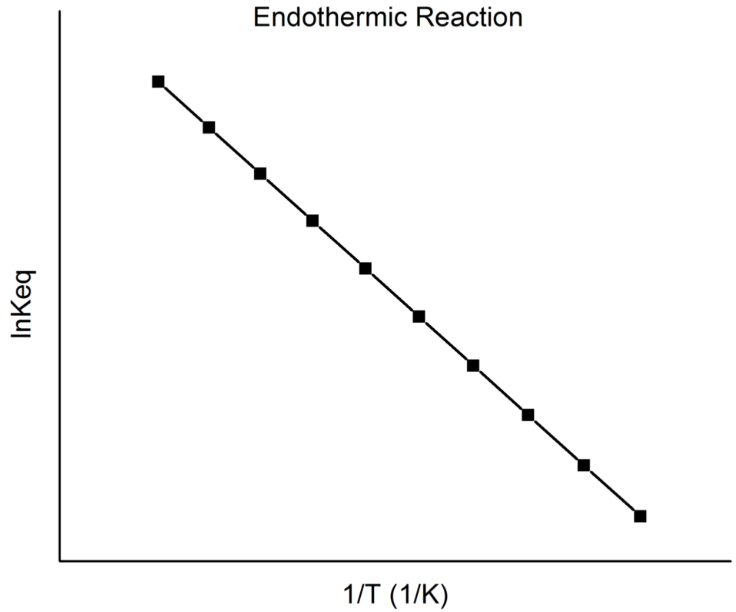
Integration leads to

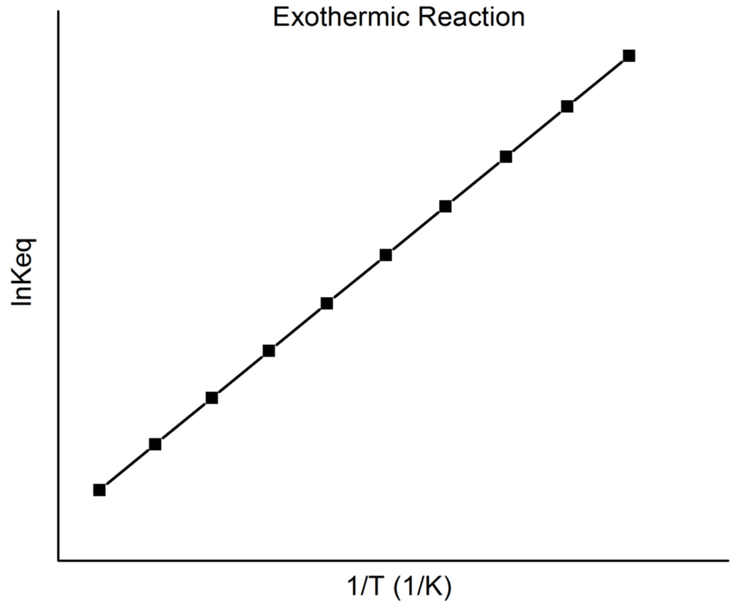
$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^{\ominus}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

Last class

T-Dependence of K

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^{\ominus}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$





(Image: https://en.wikipedia.org/wiki/Van_%27t_Hoff_equation)

Where Are We?

Foundations

- Basic concepts
- Gases
- Thermodynamics
- Statistical mechanics

Applications

- Phases
- Simple mixtures
- Chemical equilibrium
- Electrochemistry

10. Electrochemistry

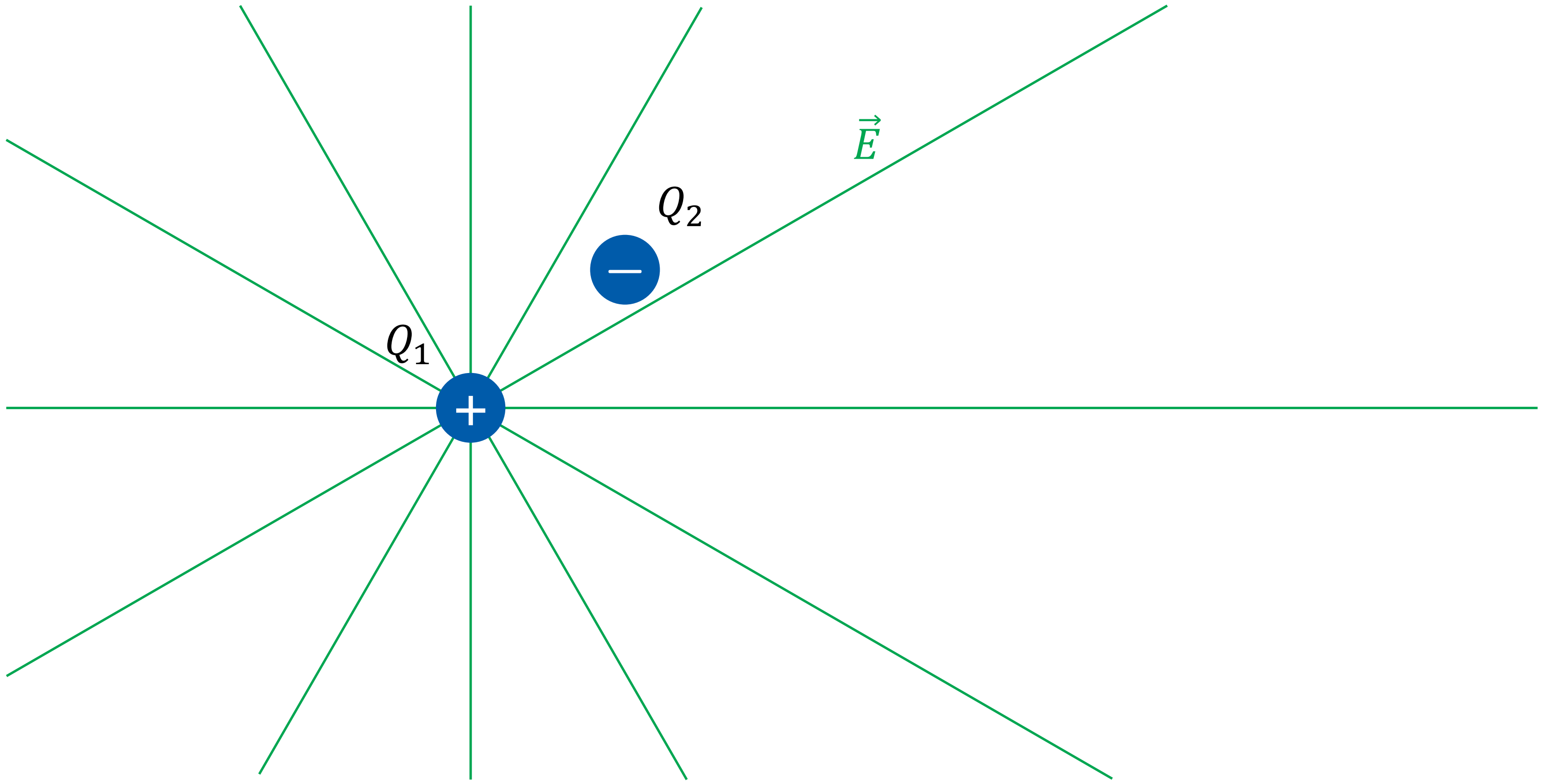
Chapter Overview

- Fundamentals of electrochemical cells
  - Review: electrostatics
  - Thermodynamics of cells
- Ionics
  - Solvation energy of ions
  - Distribution of ions
  - Activities of ions
- Electrodics
  - Electric double layer

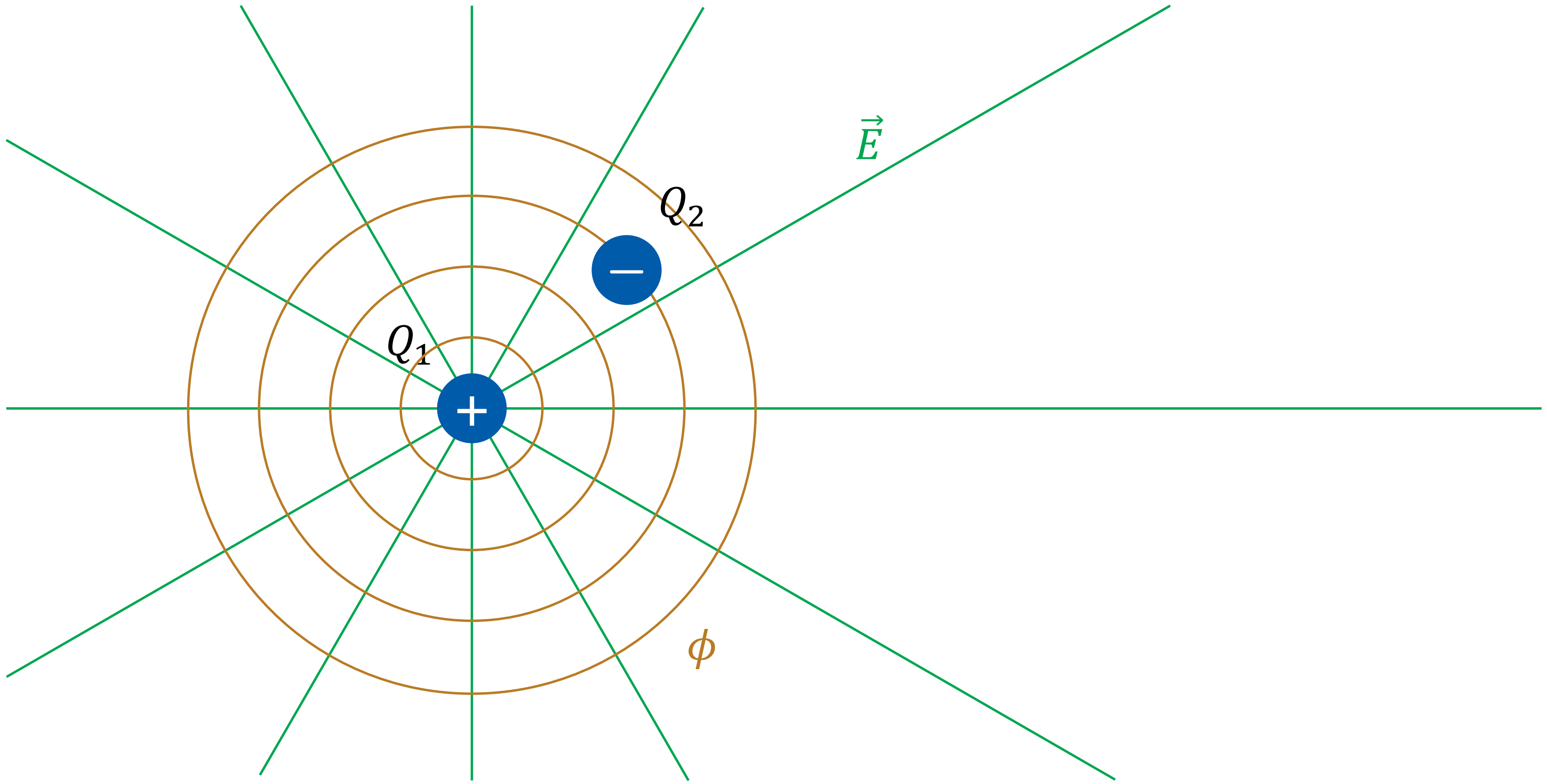
Electrostatic Force

$$\vec{F}(\vec{r}) = \frac{Q_1 Q_2}{4\pi\varepsilon r^2}\hat{r}$$

Electric Field



Electric Potential



Summary: Electrostatics

- Force (Coulomb's law)

$$\vec{F}(\vec{r}) = \frac{Q_1 Q_2}{4\pi\varepsilon r^2}\hat{r}$$

- Potential energy

$$E_\mathrm{p}(r) = \frac{Q_1 Q_2}{4\pi\varepsilon r}$$

- Electric field around Q_1

$$\vec{E} = \frac{Q_1}{4\pi\varepsilon r^2}\hat{r}$$

$$\vec{F} = Q_2\vec{E}$$

- Potential around Q_1

$$\phi(r) = \frac{Q_1}{4\pi\varepsilon r}$$

$$E_\mathrm{p} = Q_2\phi$$

Electric Work

For a charge Q in the electric potential ϕ,

$$E_{\mathrm{p}} = Q\phi$$

Consider its infinitesimal (reversible) change:

$$\delta w_{\mathrm{elec}} = \mathrm{d}E_{\mathrm{p}} = \mathrm{d}(Q\phi) = Q\mathrm{d}\phi + \phi\mathrm{d}Q$$

- If ϕ is the independent variable, $\delta w_{\mathrm{elec}} = Q\mathrm{d}\phi$
- If Q is the independent variable, $\delta w_{\mathrm{elec}} = \phi\mathrm{d}Q$

Several Types of Work

$$\delta w = \text{(intensive factor)} \times \text{(extensive factor)}$$

Type of work	dw	Comments	Units†	
Expansion	$-p_{\text{ex}}\text{d}V$	p_{ex} is the external pressure	Pa	δw_{exp}
		dV is the change in volume	m^3	
Surface expansion	$\gamma\text{d}\sigma$	γ is the surface tension	N m^{-1}	δw_{add}
		dσ is the change in area	m^2	
Extension	$f\text{d}l$	f is the tension	N	
		dl is the change in length	m	
Electrical	$\phi\text{d}Q$	ϕ is the electric potential	V	
		dQ is the change in charge	C	
	$Q\text{d}\phi$	dϕ is the potential difference	V	
		Q is the charge transferred	C	

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12^{th} ed., Table 2A.1)

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Last Generalization

For a non-expansion work $\delta w = f\mathrm{d}X$,

- State: $(p, T, n_\mathrm{A}, n_\mathrm{B}, \cdots) \rightarrow (p, T, X, n_\mathrm{A}, n_\mathrm{B}, \cdots)$

- Thermodynamic potentials
  - $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + f\mathrm{d}X + \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} + \cdots$
  - $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p + f\mathrm{d}X + \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} + \cdots$
  - $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + f\mathrm{d}X + \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} + \cdots$
  - $\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + f\mathrm{d}X + \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} + \cdots$

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Examples

- Surface tension

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \gamma\mathrm{d}\sigma + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$$

- Electric work (Q as an independent variable)

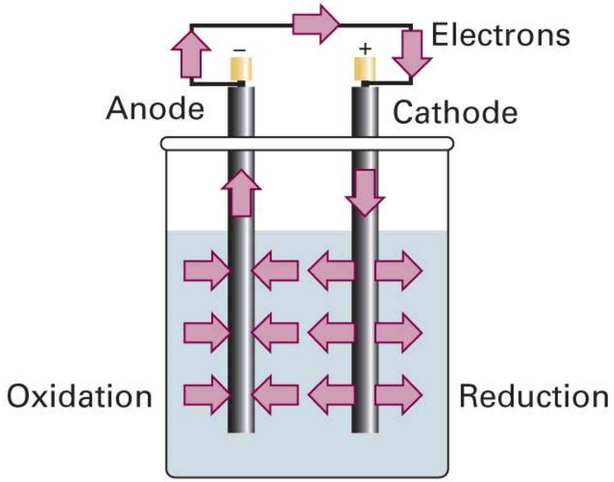
$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \phi\mathrm{d}Q + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$$

- Electric work (ϕ as an independent variable)

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + Q\mathrm{d}\phi + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$$

Electrochemical Cell

- Two electrodes + electrolytes
  - Anode: oxidation
  - Cathode: reduction



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 6C.1)

Half Cells and Cell Diagram

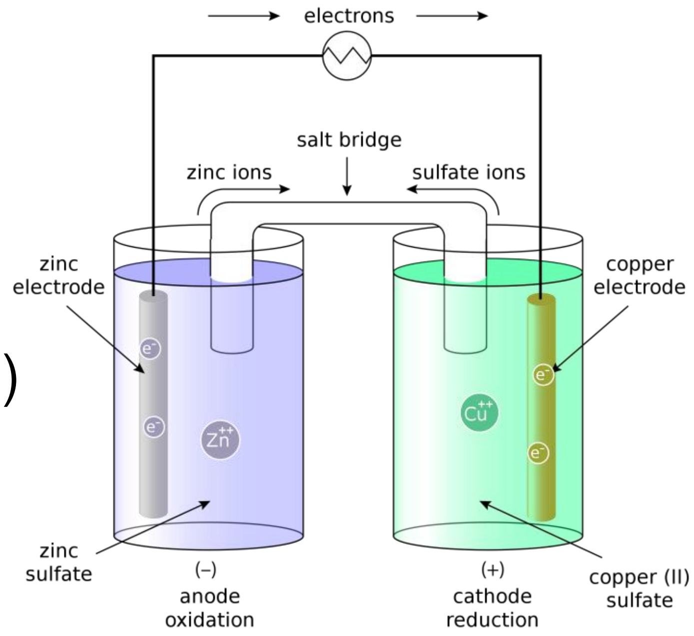
- Cell with $Zn^{2+}(aq)/Zn(s)$ and $Cu^{2+}(aq)/Cu(s)$ electrodes

anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

- Cell diagram

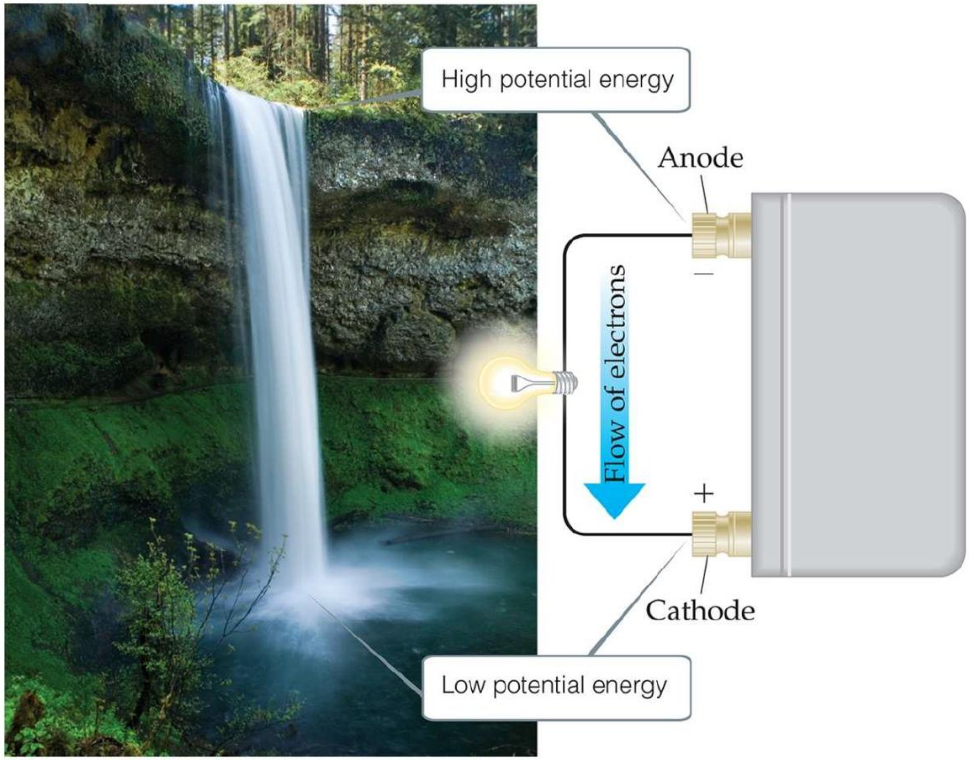
$Zn(s)|ZnSO_4(aq)||CuSO_4(aq)|Cu(s)$



(Image: https://commons.wikimedia.org/wiki/File:Galvanic_cell_labeled.svg)

Cell Potential

Potential difference between the two electrodes



(Image: https://www.unf.edu/~michael.lufaso/chem2046/2046chapter20.pdf)

Thermodynamics of Cell

Recall that

$$\mathrm{d}G_{T,p} = \delta w_{\mathrm{add,max}} = \delta w_{\mathrm{elec}}$$

for a reversible electrochemical reaction.

For an infinitesimal progress of the reaction $\mathrm{d}\xi$,

$$\mathrm{d}G = \Delta_{\mathrm{r}}G\mathrm{d}\xi$$

Hence,

$$\delta w_{\mathrm{elec}} = \Delta_{\mathrm{r}}G\mathrm{d}\xi$$

Electrical Work

$$\delta w_{\mathrm{elec}} = \phi \mathrm{d}Q$$

where ϕ is the electric field and $\mathrm{d}Q$ is the transferred charge.

For the electrochemical reaction,

$$\phi = E_{\mathrm{cell}}$$

$$\mathrm{d}Q = -\nu e N_{\mathrm{A}} \mathrm{d}\xi = -\nu F \mathrm{d}\xi$$

Hence,

$$\delta w_{\mathrm{elec}} = -\nu F E_{\mathrm{cell}} \mathrm{d}\xi$$

Thermodynamics of Cell

$$\delta w_{\mathrm{elec}} = \Delta_{\mathrm{r}} G \mathrm{d}\xi$$

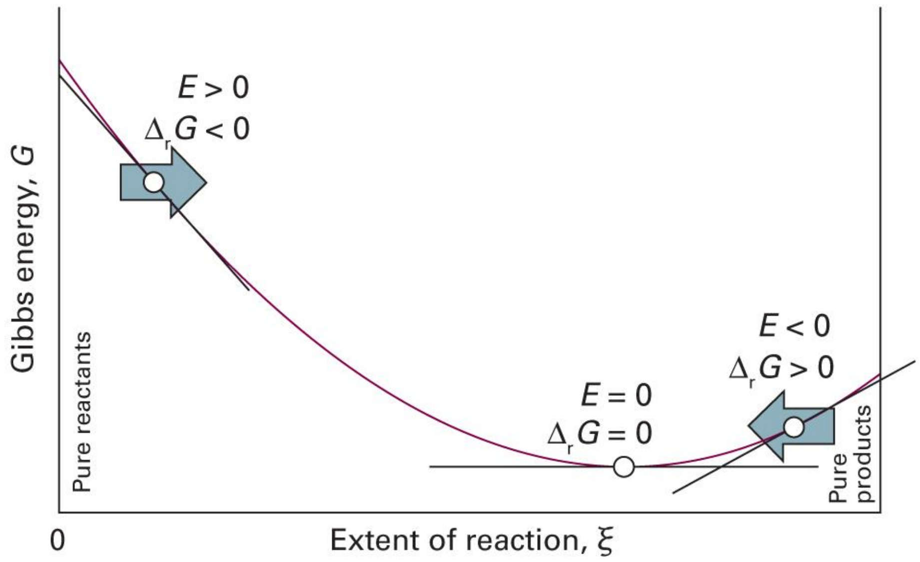
$$\delta w_{\mathrm{elec}} = -\nu F E_{\mathrm{cell}} \mathrm{d}\xi$$

Combining the two, we obtain

$$\Delta_{\mathrm{r}} G = -\nu F E_{\mathrm{cell}}$$

Cell Potential and Reaction

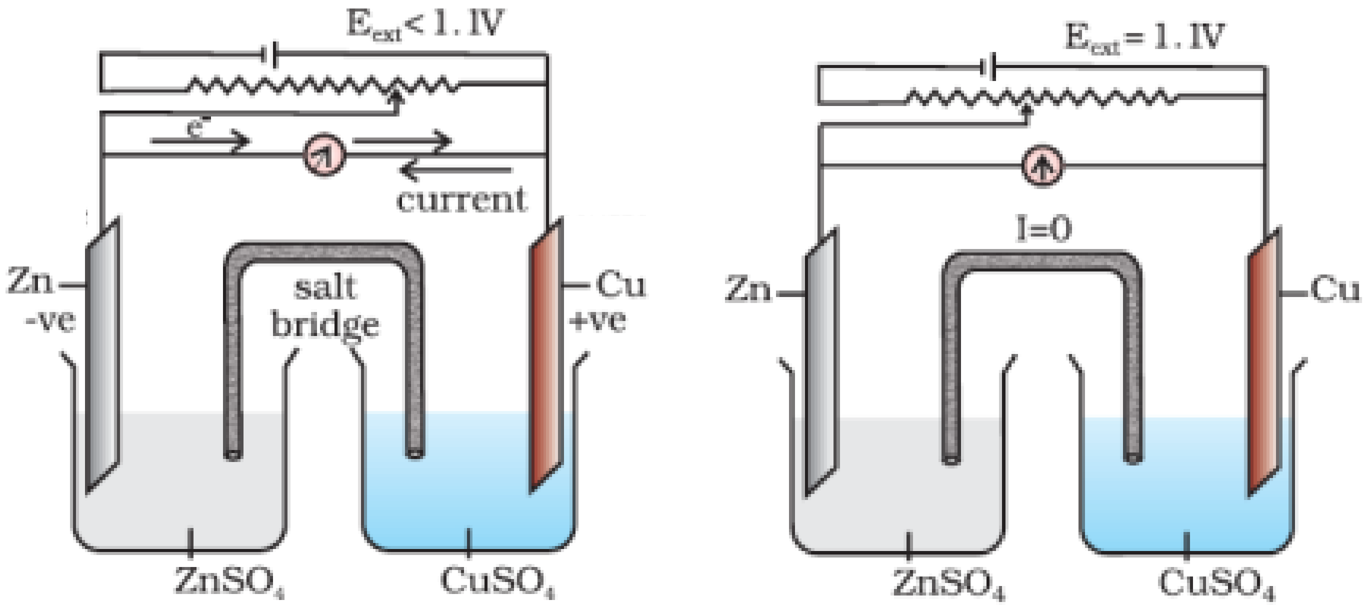
$$\Delta_r G = -\nu F E_{cell}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 6C.4)

Topic 6C

Measurement of Cell Potential



$$E_{\text{cell}} = (E_{\text{ext}})_{I=0}$$

(Image: https://ncerthelp.com/query1.php?Q=ncert+solutions+class+12+Chemistry+Chapter+3+ELECTROCHEMISTRY)

Nernst Equation

$$\Delta_r G = -\nu F E_{\text{cell}}$$

Using $\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q$,

$$-\nu F E_{\text{cell}} = \Delta_r G^{\ominus} + RT \ln Q$$

Thus,

$$E_{\text{cell}} = E^{\ominus}_{\text{cell}} - \frac{RT}{\nu F} \ln Q$$

where $E^{\ominus}_{\text{cell}} = -\Delta_r G^{\ominus}/\nu F$

At Equilibrium

$$E_{\text{cell}} = E^{\ominus}_{\text{cell}} - \frac{RT}{\nu F}\ln Q$$

At equilibrium, $E_{\text{cell}} = 0$ and $Q = K$

$$0 = E^{\ominus}_{\text{cell}} - \frac{RT}{\nu F}\ln K$$

or,

$$E^{\ominus}_{\text{cell}} = \frac{RT}{\nu F}\ln K$$

Temperature Coefficient

$$E^{\ominus}_{\text{cell}} = -\frac{\Delta_{\text{r}}G^{\ominus}}{\nu F}$$

Since $(\partial G/\partial T)_p = -S$,

$$\frac{\mathrm{d}E^{\ominus}_{\text{cell}}}{\mathrm{d}T} = -\frac{1}{\nu F}\left(\frac{\partial \Delta_{\text{r}}G^{\ominus}}{\partial T}\right)_p = \frac{\Delta_{\text{r}}S^{\ominus}}{\nu F}$$

Furthermore,

$$\Delta_{\text{r}}H^{\ominus} = \Delta_{\text{r}}G^{\ominus} + T\Delta_{\text{r}}S^{\ominus} = -\nu F\left(E^{\ominus}_{\text{cell}} - T\frac{\mathrm{d}E^{\ominus}_{\text{cell}}}{\mathrm{d}T}\right)$$

Electrode Potentials

Cell potential = cathode contribution + anode contribution

For a cell L‖R,

$$E^{\ominus}_{\text{cell}} = E^{\ominus}(\text{R}) - E^{\ominus}(\text{L})$$

Impossible to measure each contribution

Convention

- Standard hydrogen electrode (SHE): $Pt(s)|H_2(g)|H^+(aq)$

$$E^{\ominus} = 0 \text{ at all temperatures}$$

- Standard potential $E^{\ominus}(X)$

For a cell $Pt(s)|H_2(g)|H^+(aq)\|X$, $E^{\ominus}(X) = E^{\ominus}_{\mathrm{cell}}$

Standard Potentials

Couple	$E^{\ominus}/\mathrm{V}$
$Ce^{4+}(aq) + e^- \rightarrow Ce^{3+}(aq)$	+1.61
$Cu^{2+}(aq) + 2\,e^- \rightarrow Cu(s)$	+0.34
$AgCl(s) + e^- \rightarrow Ag(s) + Cl^-(aq)$	+0.22
$H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g)$	0
$Zn^{2+}(aq) + 2\,e^- \rightarrow Zn(s)$	−0.76
$Na^+(aq) + e^- \rightarrow Na(s)$	−2.71

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Table 6D.1)

Thermodynamics of Potentials

Each electrode potential can be converted into

$$-\nu F E^{\ominus} = \Delta_{\mathrm{r}} G^{\ominus}$$

When a combination of reactions a and b gives reaction c,

$$\Delta_{\mathrm{r}} G^{\ominus}(\mathrm{c}) = \Delta_{\mathrm{r}} G^{\ominus}(\mathrm{a}) + \Delta_{\mathrm{r}} G^{\ominus}(\mathrm{b})$$

$$-\nu_{\mathrm{c}} F E^{\ominus}(\mathrm{c}) = -\nu_{\mathrm{a}} F E^{\ominus}(\mathrm{a}) - \nu_{\mathrm{b}} F E^{\ominus}(\mathrm{b})$$

Thus,

$$\nu_{\mathrm{c}} E^{\ominus}(\mathrm{c}) = \nu_{\mathrm{a}} E^{\ominus}(\mathrm{a}) + \nu_{\mathrm{b}} E^{\ominus}(\mathrm{b})$$

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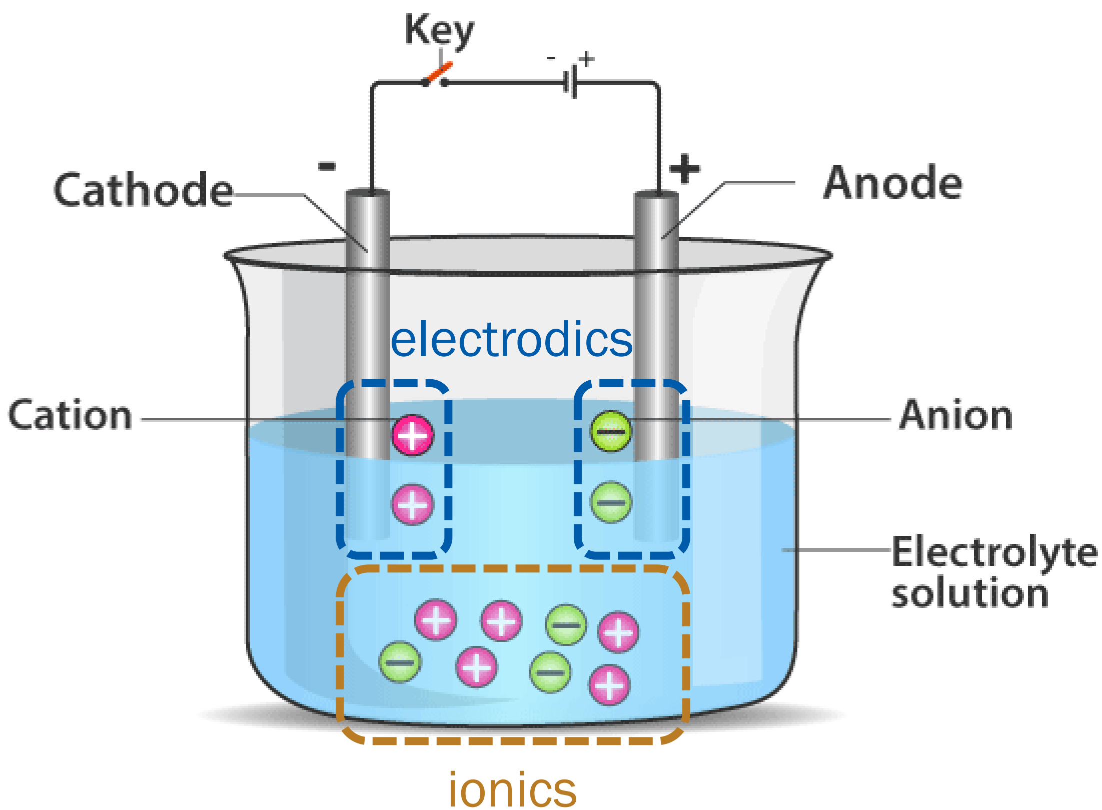
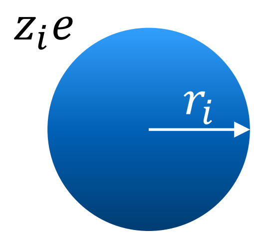
Electrochemistry



(Image: https://byjus.com/physics/electrolysis-and-electroplating/)

Work for Charging an Ion

Model each ion i as a sphere with radius r_i and charge $z_i e$.



The electric work required to charge this ion is

$$\delta w_{\text{elec}} = \phi \mathrm{d}Q$$

$$w_{\text{elec}} = \int_0^{z_i e} \phi \, \mathrm{d}Q = \int_0^{z_i e} \left(\frac{Q}{4\pi\varepsilon r_i} \right) \mathrm{d}Q = \frac{z_i^2 e^2}{8\pi\varepsilon r_i}$$

Gibbs Energy for Solvation

$$w_{\text{elec}} = \frac{z_i^2 e^2}{8\pi\varepsilon r_i}$$

Since $\Delta G = w_{\text{add,max}} = w_{\text{elec,max}}$,

$$\Delta_{\text{solv}} G^{\ominus} \approx w(\text{medium}) - w(\text{vacuum})$$

For vacuum, $\varepsilon = \varepsilon_0$ and

$$\Delta_{\text{solv}} G^{\ominus} = \frac{z_i^2 e^2}{8\pi\varepsilon r_i} - \frac{z_i^2 e^2}{8\pi\varepsilon_0 r_i} = -\frac{z_i^2 e^2}{8\pi\varepsilon_0 r_i}\left(1 - \frac{\varepsilon_0}{\varepsilon}\right)$$

Born Equation

$$\Delta_{\text{solv}}G^{\ominus} = -\frac{z_i^2 e^2}{8\pi\varepsilon_0 r_i}\left(1 - \frac{1}{\varepsilon_{\text{r}}}\right)$$

where the relative permittivity $\varepsilon_{\text{r}} = \varepsilon/\varepsilon_0$.

- $\Delta_{\text{solv}}G^{\ominus} < 0$: solvation of ions is favored
- Water has ε_{r} = 78.54 at 25°C, and

$$\Delta_{\text{solv}}G^{\ominus} = -\frac{z_i^2}{r_i/\text{pm}} \times (6.86 \times 10^4\ \text{kJ mol}^{-1})$$

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Generalized Born Model

$$\Delta_{\text{solv}} G^{\ominus} = -\frac{z_i^2 e^2}{8\pi\varepsilon_0}\left(1 - \frac{1}{\varepsilon_{\text{r}}}\right)\frac{1}{r_i}$$

We can generalize the Born model by using

$$\Delta_{\text{solv}} G^{\ominus} = -\frac{z_i^2 e^2}{8\pi\varepsilon_0}\left(1 - \frac{1}{\varepsilon_{\text{r}}}\right)\frac{1}{\sqrt{r_i^2 + a_i^2 e^{-(r_i/2a_i)^2}}}$$

where a_i is the effective Born radius.

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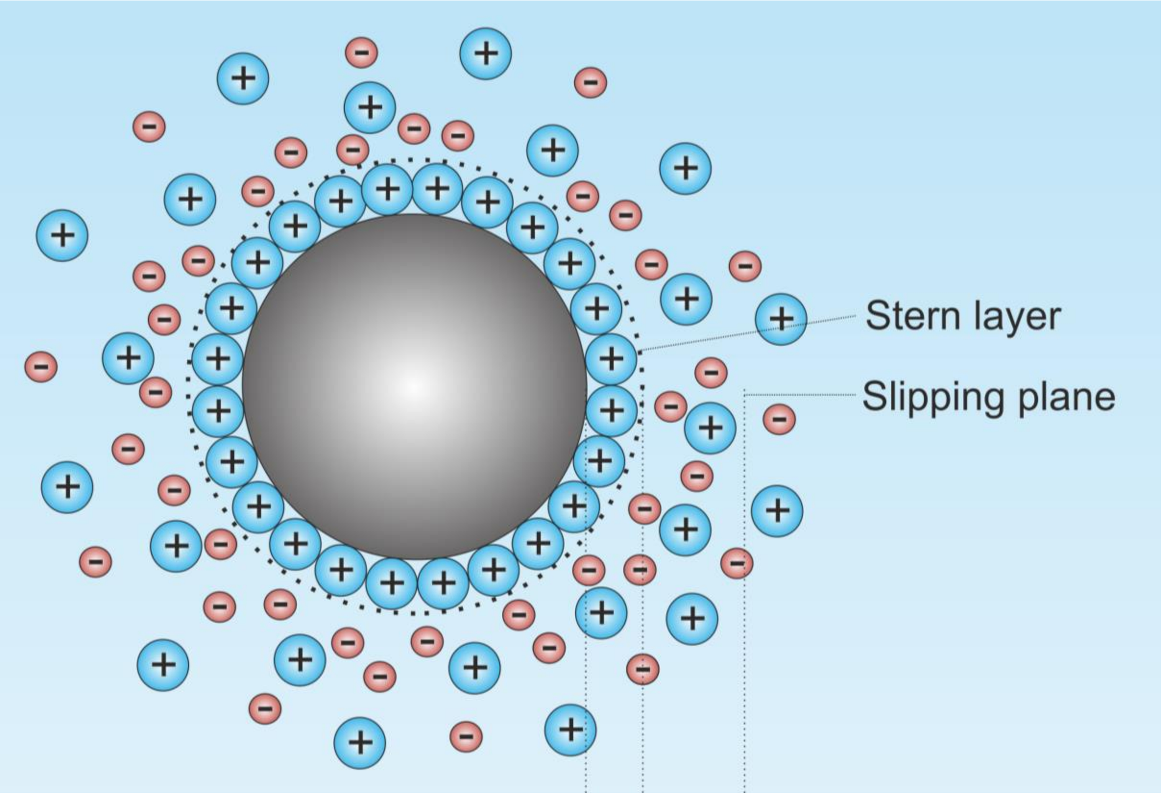
Distribution of Ions in ϕ



(Image: https://commons.wikimedia.org/wiki/File:Zeta_Potential_for_a_particle_in_dispersion_medium.png)

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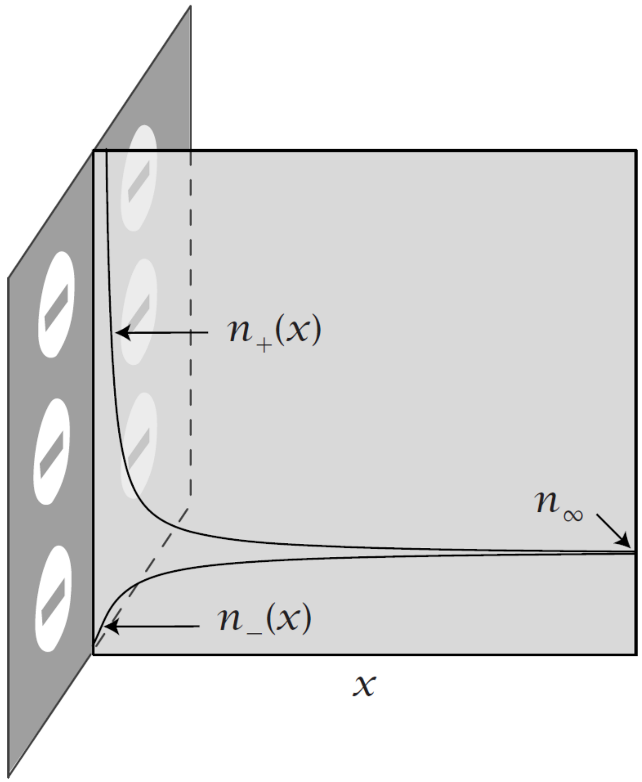
Boltzmann Distribution



In the electric potential ϕ,

- Potential energy

$$E_\mathrm{p} = Q\phi$$

- Boltzmann distribution

$$P \propto e^{-E_\mathrm{p}/k_\mathrm{B}T} = e^{-Q\phi/k_\mathrm{B}T}$$

- Mole numbers of ions

$$n_+ = n_\infty e^{-ze\phi/k_\mathrm{B}T}$$

$$n_- = n_\infty e^{+ze\phi/k_\mathrm{B}T}$$

(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figure 23.2)

Review

Poisson Equation

Relationship between charge density and potential

For an arbitrary charge density $\rho(\vec{r})$, the potential $\phi(\vec{r})$ satisfies

$$\nabla^2\phi = -\frac{\rho}{\varepsilon}$$

(Laplacian $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$)

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Poisson-Boltzmann Equation

- Charge density

Faraday constant eN_A

$$\rho = zeN_\mathrm{A}n_+/V + (-ze)N_\mathrm{A}n_-/V = zF(n_+ - n_-)/V$$

$$= zFn_\infty\left(e^{-ze\phi/k_\mathrm{B}T} - e^{+ze\phi/k_\mathrm{B}T}\right)/V$$

- Poisson equation

$$\nabla^2\phi = -\rho/\varepsilon$$

- Poisson-Boltzmann equation

$$\nabla^2\phi = \frac{zFn_\infty}{\varepsilon V}\left(e^{+ze\phi/k_\mathrm{B}T} - e^{-ze\phi/k_\mathrm{B}T}\right)$$

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Debye-Hückel Equation

$$\nabla^2\phi = \frac{zFn_\infty}{\varepsilon V}\left(e^{+ze\phi/k_\mathrm{B}T} - e^{-ze\phi/k_\mathrm{B}T}\right)$$

At the limit of a dilute solution, $ze\phi/k_\mathrm{B}T \ll 1$ and

$$e^{\pm ze\phi/k_\mathrm{B}T} \approx 1 \pm ze\phi/k_\mathrm{B}T = 1 \pm zF\phi/RT$$

Hence,

$$\nabla^2\phi = \frac{zFn_\infty}{\varepsilon V}\{(1 + zF\phi/RT) - (1 - zF\phi/RT)\}$$

$$\nabla^2\phi = \frac{2z^2F^2n_\infty}{\varepsilon VRT}\phi$$

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Ion Strength

$$I = \frac{1}{2}\sum_i z_i^2 \left(\frac{b_i}{b^\ominus}\right)$$

For two types of ions with $|z_+| = |z_-| = z$,

$$I = \frac{z_+^2 b_+ + z_-^2 b_-}{2b^\ominus} = \frac{z^2 b}{b^\ominus}$$

Since $b = n_\mathrm{B}/m_\mathrm{A} = n_\infty/m_\mathrm{A}$,

$$I = \frac{z^2 n_\infty}{b^\ominus m_\mathrm{A}}$$

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Generalization of D-H Eq.

$$\nabla^2\phi = \frac{2F^2z^2n_\infty}{\varepsilon VRT}\phi$$

Since the ion strength is $I = z^2n_\infty/b^{\ominus}m_{\mathrm{A}}$,

$$\nabla^2\phi = \frac{2F^2Ib^{\ominus}(m_{\mathrm{A}}/V)}{\varepsilon RT}\phi$$

This equation is generally applicable to any salt system.

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Debye Length

$$r_{\mathrm{D}} = \sqrt{\frac{\varepsilon RT}{2F^2 I b^{\ominus}(m_{\mathrm{A}}/V)}} \propto \frac{1}{\sqrt{I}}$$

Then

$$\nabla^2\phi = \phi/r_{\mathrm{D}}^2$$

For an aqueous solution at room temperature,

$$r_{\mathrm{D}} \approx \left(0.304\ \mathrm{nm}\ \mathrm{M}^{1/2}\right)/\sqrt{I}$$

Not in Atkins

Screening Effect

For a point charge of Q, solving the differential equation

$$\nabla^2\phi = \phi/r_{\mathrm{D}}^2$$

gives a solution of

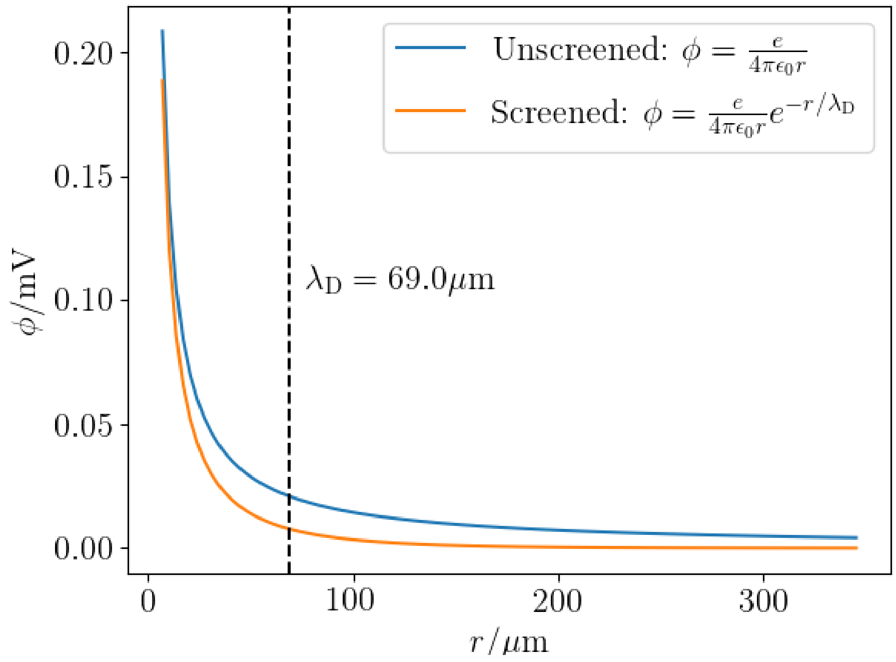
$$\phi(r) = \frac{Q}{4\pi\varepsilon r}e^{-r/r_{\mathrm{D}}}$$

which can be compared to the Coulombic potential

$$\phi(r) = \frac{Q}{4\pi\varepsilon r}$$

Not in Atkins

Screening Effect



(Image: https://scipython.com/blog/the-debye-length)